

Keysight SalesOps — Use-case coverage matrix

Generated 2026-05-13 · audits every intent × gate × edge-case path the demo currently exercises.

This is the single-source reference the Keysight review team can use to confirm what the SalesOps automation does end-to-end, where the autonomy gates sit, and where every edge case lands. It is built from the live pipeline taxonomy, the four-gate confidence model, the AIOA webhook handoff, and the per-intent action matrix. Each row names the trigger, the gates the case passes through, the action ZBrain proposes, the downstream system that owns the work, and the failure / HITL path.

1. Pipeline-level coverage

Stage	Sub-process	What ZBrain does	Tier outcomes	Owning system
Pre-Intake (Outlook rules)	Bounce / DSN detection	Deterministic regex on subject / from / body. Skips all six stages.	L4 discard	ZBrain
	Out-of-office reply	Pattern match on OOO phrases in EN/ES/JA.	L4 discard	ZBrain
	Brazil tax / Collections / Portal admin / KSO	Keyword + sender allowlist; routes to named team.	L4 redirect (CommunicationLog only)	Keysight team (forward)
	Fall-through	No rule matched — continues to Stage 1 LLM classifier.	n/a	ZBrain
Stage 1 — Intake & Classification	Language detect	EN / ES / JA detection on inbound body.	Always succeeds (defaults to EN)	ZBrain
	Intent classification (LLM, two-pass)	Context-pass + override-pass; CSR-typed do-not-auto / force-HITL recognised.	L4 / L3 / L2 governed by Gate 1 confidence	ZBrain
	Spam / phishing screen	LLM screen + heuristic redundancy.	L4 discard on agreement	ZBrain
	Multi-intent split	Sub-intents recorded for downstream branching (po_intake + wo_status_inquiry, etc.).	Primary drives action; secondary surfaces in trace	ZBrain
Stage 2 — Extraction & Enrichment	OCR / document intelligence	Vision OCR for PDFs / XLSX / DOCX / images.	Gate 2 (extraction)	ZBrain
	Intent-aware schema extraction	Required fields per intent driven by KB extract_schema namespace.	Gate 2 (extraction)	ZBrain
	Salesforce customer match	Full SOQL match using extracted JSON; local-DB fallback for the demo when no SF connection.	Gate 3 (entity resolution)	Salesforce
	CMD activation request	When SF customer not found, emit "CMD activation needed" event for the operator.	L2 HITL — unknown_customer_in_salesforce	Salesforce CMD
	Cross-system reconcile (Stage 2.5)	PO ↔ matched quote line-item check.	Findings feed Action Feasibility (Gate 4)	ZBrain
Stage 3 — Decision & Confidence	Existing-CCC lookup (3.0a)	SF Case lookup by PO# / WO# — routes to update / clone change order / new.	Branches downstream	Salesforce
	Routing resolver (3.0b)	Distributor + magic-SKU rules pick the right queue.	Routing_target attached	ZBrain

Stage	Sub-process	What ZBrain does	Tier outcomes	Owning system
	AIOA webhook handoff (3.0c)	POST to Keysight AIOA endpoint for po_intake / quote_to_order / trade_change_order / wo_update_request / service_contract_request when a PO is attached.	AIOA owns validation + downstream work; ZBrain records outcome	AIOA (external)
	Four-gate confidence + business rules (3.1–3.3)	Composite = min(Classification, Extraction, Entity Resolution, Action Feasibility); KB rule caps applied; CSR override forces L2_HITL.	L4 ≥ 0.95 / L3 0.80–0.94 / L2 < 0.80	ZBrain
Stage 4 — Workflow Execution	AIOA handoff short-circuit	If AIOA fired, ZBrain skips its own SF Order write — AIOA owns it.	handed_off_to_aioa / aioa_fallout_queue	AIOA
	Customer-match guardrail (4.1)	Refuse SF write when Stage 2.3 didn't resolve an Account.	L2 HITL	Salesforce
	Idempotency check (4.2)	SOQL lookup for existing Order with same PoNumber on the Account.	Skip duplicate write; mark idempotent	Salesforce
	Workflow execution (4.3)	Per-action write: Salesforce Order / WO / Status update / Contract draft.	L4 applies, L3 stages draft + one-click, L2 HITL preview	Salesforce (Oracle EBS / DocuNet via Jitterbit is upcoming)
Stage 5 — Communication & Close-out	Draft customer reply (5.1)	LLM-drafted reply in customer language; deterministic template fallback if the LLM fails.	Always produces a draft	ZBrain
	Translate to customer language (5.2)	Skipped when language is EN; otherwise EN preview generated for CSR review.	Tool-result feed	ZBrain
	Attach SOA / file in SharePoint (5.3)	SOA PDF generated for po_intake / quote_to_order / trade_change_order. Upload to SharePoint when active connection; DocuNet handoff queued if the placeholder integration is enabled.	L4 autosend / L3 awaiting click / L2 HITL preview	SharePoint (DocuNet via Jitterbit upcoming)
	Communication log (5.4)	CommunicationLog row written on L4_AUTO; outbound SMTP send (demo lock blocks the actual send).	Visible in Dashboard → Autonomous replies (L4)	ZBrain
Stage 6 — Continuous Learning	Signal capture (Loop A)	CSR 👍 / 👎 / edit feedback persisted on the trace.	Feedback rows	ZBrain
	Opportunity / promotion (Loops B + C)	Drift detection, A/B-promoted KB rules, operator-tunable thresholds.	Surfaced on /learning	ZBrain
	Drift & anomaly detection (Loop E)	Per-intent confidence drift watchdog.	Alerting hooks	ZBrain

2. Intent × action × downstream system

Intent	Primary action	Action_required fields the CSR signs off	Downstream systems	AIOA webhook?	SOA generated?
po_intake	create_order_acknowledgment	customer_name, po_number, quote_number, currency, total, line_item_count, requested_ship_date, ship_to	Salesforce Order, SharePoint (SOA), customer email	Yes (when PO attached)	Yes
quote_to_order	convert_quote_to_order	customer_name, quote_number, currency, total	Salesforce Order, customer email	Yes (when PO attached)	Yes
trade_change_order	trade_change_order	source_case_id, change_summary, po_reference	Salesforce Case clone, Salesforce Order (CO), customer email	Yes (when PO attached)	Yes
ssd_change_request	reschedule_order	order_number, new_ship_date, reason	Salesforce Order, customer email	No	No
hold_release	release_hold	order_number, hold_type, release_reason	Salesforce Order, customer email	No	No
delivery_change	reschedule_order	order_number, new_ship_date	Salesforce Order, customer email	No	No
service_order	create_work_order	customer_name, service_type, site_location, asset_count, requested_completion_date, po_reference	Salesforce Service WorkOrder, field dispatch, customer email	No	No
wo_update_request	update_work_order	work_order_number, update_instruction	Salesforce WorkOrder, customer email	Yes (when PO attached)	No
wo_status_inquiry	answer_wo_status	work_order_number, current_status	Customer email only	No	No
service_contract_request	create_service_contract	customer_name, contract_term_years, annual_value, currency, coverage_summary	Salesforce contract draft, finance review, customer email	Yes (when PO attached)	No
general_inquiry	(reply only)	—	Customer email	No	No
kso / collections / portal_admin / brazil_tax	redirect (Pre-Intake)	—	Forwarded to named team (demo lock blocks SMTP)	No	No
out_of_scope / spam / undeliverable	discard	—	None	No	No

3. Edge-case paths the demo exercises

Edge case	Where it surfaces	What happens	UI surface
AIOA Pass → automation complete	Stage 3.0c → Stage 4 short-circuit	ZBrain trace records the handoff; downstream order acceptance is owned by AIOA. No SF Order write from ZBrain.	Trace, Dashboard AIOA tile
AIOA Fail → fallout queue	Stage 3.0c → Stage 4 short-circuit	Case routed to AI OA Fallout queue (in AIOA's surface). ZBrain records the reason.	Trace, Dashboard AIOA tile
Unknown customer (CMD activation needed)	Stage 2.3 entity resolution gate	Pipeline drops to L2 HITL with reason <code>unknown_customer_in_salesforce</code> . CMD activation event emitted.	HITL queue, Dashboard CMD tile
Magic SKU (CUSTOM-PRODUCT, SOWDUMMY, EXPORTDUMMY)	Routing resolver (3.0b) + AIOA findings	Routes to engineering / SOW / export-compliance queue depending on the SKU.	Trace, routing target field
Multi-asset fan-out	<code>service_order</code> with multiple Model/Serial pairs	One CCC Case per asset.	Trace + <code>execution.applied.work_orders</code>
Two-pass classifier override	Stage 1 override-pass	LLM context pass + an explicit override pass; visible in trace if it flipped the intent.	Trace intake substeps
CSR-typed override (<code>do_not_auto</code> / <code>force_hitl</code> / <code>route_to_team</code>)	Stage 3 final tier decision	Tier forced to L2_HITL regardless of computed confidence; the typed instruction stored on the cap record.	Trace, Decision panel
Thread walk-back (10–40 message threads)	Stage 1 + Stage 2	Thread context carried back so classification + extraction use the latest customer state.	Trace, thread view
Mailbox-door triage (Outlook rules)	Pre-Intake	~25% of mail short-circuited without an LLM call; Dashboard tile shows the savings.	Dashboard mailbox-door tile
Idempotent SF Order detection	Stage 4.2	Duplicate PO write is skipped; trace surfaces the existing OrderNumber.	Trace 4.2 substep
Reconcile blocking finding	Stage 2.5 → Stage 3 Gate 4 cap	Action Feasibility gate caps at 0.30 (<code>hard_block</code>); displayed as L2_HITL with the reason.	Trace 3.2, HITL preview
Region overlay (AMS / EMEA / APAC / JP)	Stage 1 + KB region rules	Per-region intent filters and routing rules; visible in trace as guardrails.	Trace guardrails list
Stage 5 LLM failure	Stage 5.1	Deterministic template draft generated in customer language; trace flags <code>template_fallback=true</code> .	HITL preview, Trace 5.1
SOA SharePoint upload	Stage 5.3	Uploaded to <code>/SalesOps/SOA</code> when SharePoint is connected; otherwise kept in local outputs/ with <code>simulated=true</code> .	Trace 5.3 sharepoint block
DocuNet handoff queued (placeholder enabled)	Stage 5.3	When DocuNet placeholder is enabled in Settings → Integrations, the trace records the queued FCNV file with the configured endpoint.	Trace 5.3 docunet block
Existing-CCC update / clone change order	Stage 3.0a → Stage 4.0a/4.0b	Email attached + Chatter notify + status flip (update) or clone scaffold (closed cases).	Trace 4.0a / 4.0b

4. Gate-by-gate failure routing

Gate	What it measures	Fail mode	Action when capped
Gate 1 — Classification	Intent confidence from the LLM (with override-pass)	Low confidence (< 0.80)	Composite drops; tier defaults to L2 HITL
Gate 2 — Extraction	Fraction of KB-required fields populated	Required fields missing	Composite drops; <code>missing_required_fields</code> exposed on the decision
Gate 3 — Entity Resolution	Salesforce customer match score	No match / weak match	L2 HITL with <code>unknown_customer_in_salesforce</code> ; CMD activation event
Gate 4 — Action Feasibility	Can the action actually execute? Includes reconcile findings + AIOA outcome	Blocking reconcile finding / AIOA_FAIL / hard_block business rule	Composite capped at 0.30 (deep L2); reason quoted on the trace

5. What ZBrain does NOT own

- **AIOA validation work** — ZBrain triggers the webhook with the extracted PO context; AIOA executes the rule book and reports back. ZBrain never replicates AIOA's checks; the local mock simulates the round-trip for the demo.
- **Oracle EBS write** — owned by AIOA today (when AIOA fires) and by Jitterbit for the non-AIOA flows once the middleware bridge is enabled from Settings → Integrations.
- **DocuNet filing** — handed off to Keysight's document-classification overlay through Jitterbit. SharePoint is the active document store today; DocuNet is enabled from Settings → Integrations when the bridge is provisioned.
- **ServiceNow case lifecycle** — owned by the Keysight dev team. ZBrain integration is read-only / write-on-handshake; we do not run Stage 4 substeps for ServiceNow.
- **Outbound SMTP send (under demo lock)** — replies are drafted, attached, and logged as CommunicationLog rows; the actual SMTP transmission is suppressed during the demo to keep mailbox state stable.

6. Where to look in the UI

Surface	What it shows
Dashboard	Throughput, mailbox-door triage, AIOA webhook outcomes, CMD activation, four-gate confidence, autonomous replies (L4).
Inbox → Trace	Per-pipeline step-by-step with every substep, every guardrail, every tool result.
HITL queue	Per-case proposed action with the structured <code>action_required</code> block (prefilled fields, downstream systems, next-steps checklist).
Analytics → End-to-end case flow	Per-stage Celonis-style sub-process diagram + top-paths panel showing the highest-volume sequences.
Analytics → Pipeline diagnostics	Compact tiles for stage timing / tool invocations / KB rule fires / normalizer; click to expand.
Analytics → AI infrastructure cost	Real CostEvent metering of every LLM / OCR / translation call, attributed to the pipeline that produced it.
Settings → Integrations	Salesforce, ServiceNow, SharePoint, mailboxes, and the placeholder integrations (Jitterbit, DocuNet) that can be enabled with operator-supplied endpoints.
Continuous Learning	CSR feedback, drift detection, A/B-promoted KB rule updates.